Australian Math Olympiad 2020 — P2/4

Jonathan Kasongo

September 26, 2025

Australian Math Olympiad 2020 — P2/4

Amy and Bec play the following game. Initially, there are three piles, each containing 2020 stones. The players take turns to make a move, with Amy going first. Each move consists of choosing one of the piles available, removing the unchosen pile(s) from the game, and then dividing the chosen pile into 2 or 3 non-empty piles. A player loses the game if they are unable to make a move.

Prove that Bec can always win the game, no matter how Amy plays.

Solution

The following claim is the heart of the problem.

Claim 1: If, on Bec's turn, there is a pile of x stones, where $x \not\equiv 1 \pmod 3$ then, in the subsequent turns she is able to force Amy to always choose a pile of $y < x$ stones, where $y \equiv 1 \pmod 3$.

Proof: We have two cases,

- If $x \equiv 0 \pmod 3$, then Bec can split the pile of x stones like so: $1,\ 1,\ x - 2$. Each of these numbers is congruent to 1 mod 3.
- If $x \equiv 2 \pmod 3$, then Bec can split the pile of x stones like so: $1,\ x - 1$. Each of these numbers is congruent to 1 mod 3.

Thus on Amy's next turn she must pick a pile with y stones, with $y \equiv 1 \pmod 3$. Also since Amy can never choose a pile of stones with 1 stone, it must be the case that $y \in \{x - 2, x - 1\}$ and so $y < x$. Now when Amy makes her move, she will split the pile such that there is at least one new pile with $z \not\equiv 1 \pmod 3$ stones. Suppose, FTSOC, that there isn't such a pile.

- If Amy splits the pile of y stones into 2 piles with a, b stones such that $a, b \equiv 1 \pmod 3$, then $2 \equiv a + b = y \equiv 1 \pmod 3$, a contradiction.
- If Amy splits the pile of y stones into 3 piles with a, b, c stones such that $a, b, c \equiv 1 \pmod 3$, then $0 \equiv a + b + c = y \equiv 1 \pmod 3$, a contradiction.

Thus after this point, Bec can always choose a pile of x stones where $x \not\equiv 1 \pmod 3$ and in the subsequent turns she is also able to force Amy to always choose a pile of y stones, where $y \equiv 1 \pmod 3$. $\square$

Since $2020 = 673 \cdot 3 + 1 \equiv 1 \pmod 3$. Due to claim 1, Bec has a strategy to force Amy to always choose piles of stones with $k \equiv 1 \pmod 3$, whilst she can always avoid choosing piles of stones with $m \not\equiv 1 \pmod 3$. Again due to claim 1, that means that Amy will eventually have to pick a pile of stones with 1 stone in that pile, but since it is impossible to divide this pile into 2 or 3 non-empty piles, Bec will win. $\blacksquare$